

4.7 THERMAL POLLUTION

Definition

Thermal pollution is defined as the addition of excess of undesirable heat to water that makes it harmful to man, animal or aquatic life or otherwise causes significant departures from the normal activities of aquatic communities in water.

4.7.1 Sources (causes) of Thermal pollution

The following sources causes thermal pollution.

- (a) Nuclear power plants.
- (b) Coal-fired power plants.
- (c) Industrial effluents.
- (d) Domestic sewage.
- (e) Hydro-electric power.

(a) Nuclear power plants: Nuclear power plants including drainage from hospitals, research institutes, nuclear experiments and explosions, discharged a lot of unutilized heat and traces of toxic radio nuclides into nearby water streams. Emissions from nuclear reactors and processing installation are also responsible for increasing the temperature of water bodies. The operations of power reactors and nuclear fuel processing units

constitute the major contributor of heat in the aquatic environment.

Heated effluents from power plants are discharged at 10°C , higher than the receiving water which affect the aquatic flora and fauna.

(b) Coal-fired power plants: Some thermal power plants utilize coal as fuel, while a few plants use nuclear fuel. Coal-fired power plants constitute the major source of thermal pollutants. Their condenser coils are cooled with water from nearby lake or river and discharge the hot water back to the stream increasing the temperature of nearby water to about 15°C . The heated effluents decrease the dissolved oxygen content of water. It results into killing of fish and other marine organisms.

(c) Industrial effluents: Industries generating electricity like coal powered and nuclear powered plants, require huge amounts of cooling water for heat removal. Other industries like textile, paper and pulp as well as sugar also release heat in water but to a much lesser extent. Normally the discharged water from steam-electric power industry using turbo generators, will have a higher temperature ranging from 6°C to 9°C than the receiving water. This results in the increase of stream temperature to a level at which natural dissipation of heat will be inefficient.

In modern stations, producing 100 MW, nearly one million gallons are discharged in an hour with increase in temperature of the cooling water passing by 8 to 10°C .

(d) Domestic Sewage: Domestic sewage is commonly discharged into rivers, lakes, canals or streams with or without waste treatment. The municipal sewage normally has a higher temperature than the receiving water. With the increase in temperature of the receiving water, the dissolved oxygen content decreases and the demand of oxygen increases. Hence

the anaerobic condition will set up resulting in the release of foul and offensive gases in water. The marine organisms which depend on the dissolved oxygen will die out.

(e) **Hydroelectric Power:** Generation of hydroelectric power, sometimes, results in negative thermal loading in water systems. Apart from electric power industries, various factories with cooling requirement contribute to thermal loading. It has been reported that about 18% more heat is given to cooling waters in nuclear power plants than any other plant of equivalent size.

4.7.2 Introduction of thermal pollution into streams by human activities

1. Industries and power plants use water to cool machinery and then discharge the warmed water into a stream.
2. Stream (water) temperature rises when trees and tall vegetation providing shades are cut down.
3. Soil erosion caused by construction.
4. Removal of stream side vegetation.
5. Poor farming practices.

4.7.3 Effects of Thermal Pollution

1. **Reduction in dissolved oxygen:** Concentration of dissolved oxygen (DO) decreases with increase in temperature of water.
2. **Increase in Toxicity:** The rising temperature increases the toxicity of the poison present in water. A 10°C rise in temperature doubles the toxic effect of potassium cyanide, while a 80°C rise in temperature triples the toxic effects of O-xylene causing massive mortality of fish.
3. **Interference with biological activities:** Temperature is considered to be of vital significance to physiology metabolism

and biochemical process in controlling respiratory rate, digestion, excretion and overall development of aquatic organisms. The temperature changes totally disrupt the entire ecosystem.

4. Interference with reproduction: In fishes, several activities like nest building, spawning, hatching, migration and reproduction etc., depend on optimum temperature.

5. Direct mortality: Unutilized heat in water is responsible for direct mortality of aquatic organisms. The increase in temperature exhausts the micro-organisms and shortens their life span. Above a particular temperature, death occurs to fish due to failure in respiratory system, nervous system process.

6. Food storage for fish: Change in temperature alters the seasonal variation in the type and abundance of lower organisms. The fish may lack the right food at the right time.

4.7.4 Control measures (or) management of thermal pollution

The following methods can be adopted to control the high temperature caused by thermal dischargers.

1. Cooling towers: The use of water from water systems for cooling purposes, with subsequent return to the water way after passage through the condenser, is termed as cooling process. To make it more effective, cooling towers are designed to control the temperature of water. Cooling towers transfer some of the heat from hot water to the surrounding atmosphere by the process of evaporation. Cooling tower is generally used to dissipate the recovered waste heat to eliminate the problems of the thermal pollution. Cooling tower are of two types.

(a) Wet Cooling tower: Hot water, coming out from the condenser (reactor) is allowed to spray over baffles. Cool air, with high velocity, is passed from sides, which takes away the heat and cools the water.

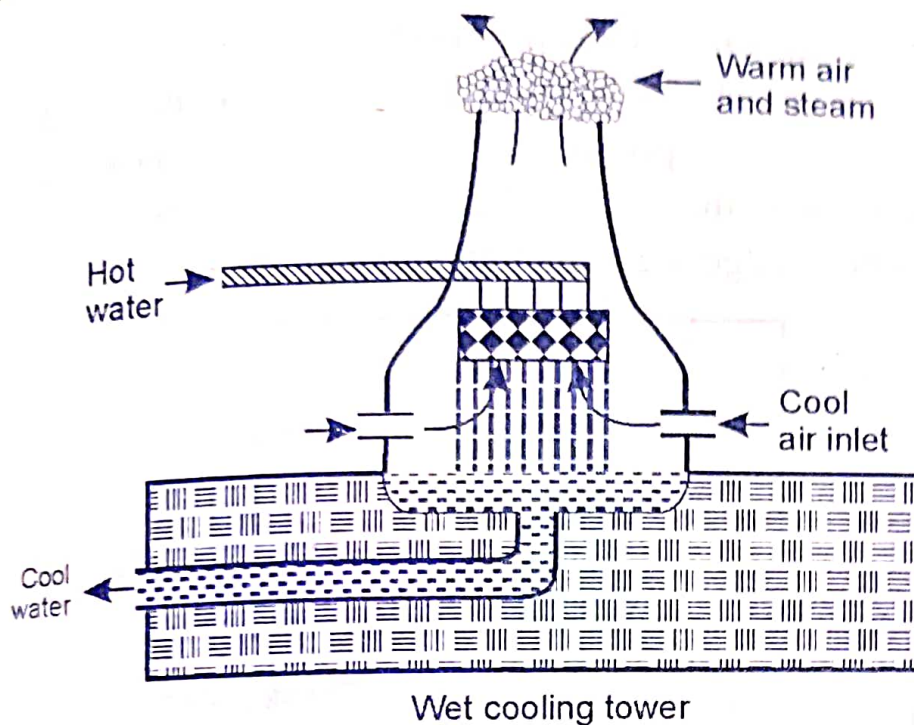


Fig. 4.4 Wet cooling tower

(b) **Dry Cooling tower:** Here the hot water is allowed to flow in a long spiral pipes. Cool air, with the help of fan, is passed over these hot pipes, which cools down the hot water. This cool water can be recycled.

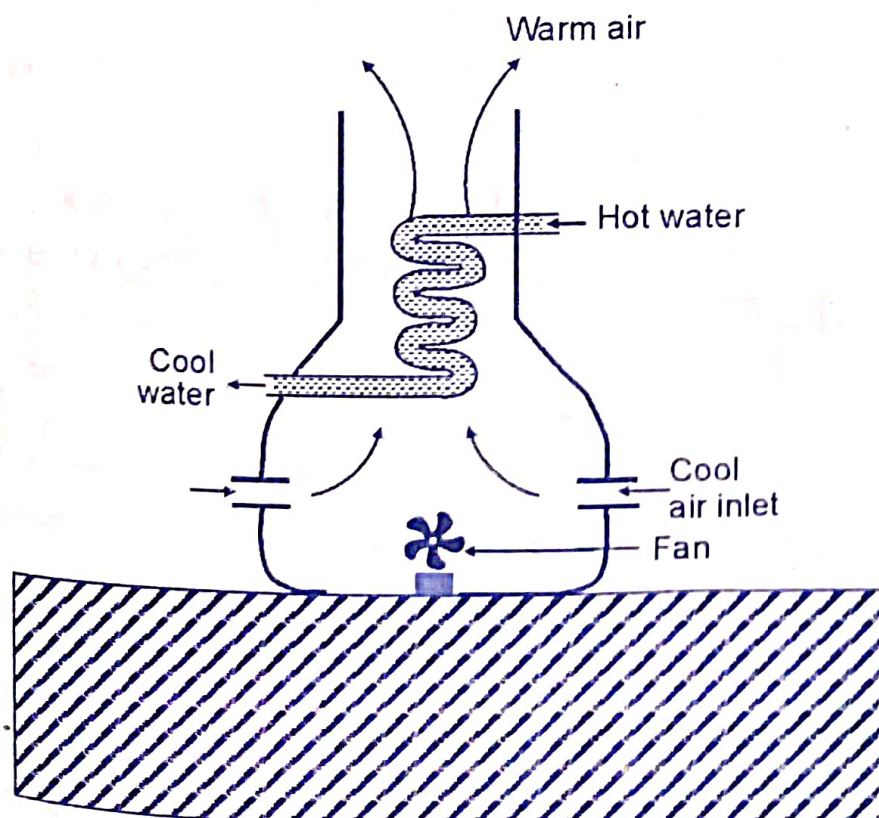


Fig. 4.5 Dry cooling tower

2. Cooling ponds: Cooling ponds are the simplest method of cooling thermal discharges. Heated effluents on the surface of water in cooling ponds maximize dissipation of heat to the atmosphere and minimize the water area and volume. The warm water wedge acts like a cooling pond.

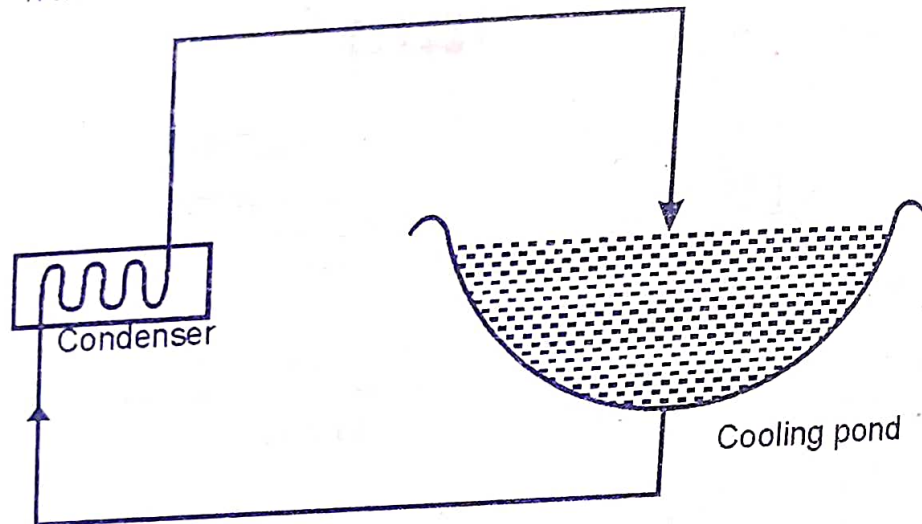


Fig. 4.6 Cooling pond

3. Spray ponds: The water coming out from condensers is allowed to pass into the ponds through sprayers. Here the water is sprayed through nozzles as fine droplets. Heat from the fine droplets gets dissipated to the atmosphere.

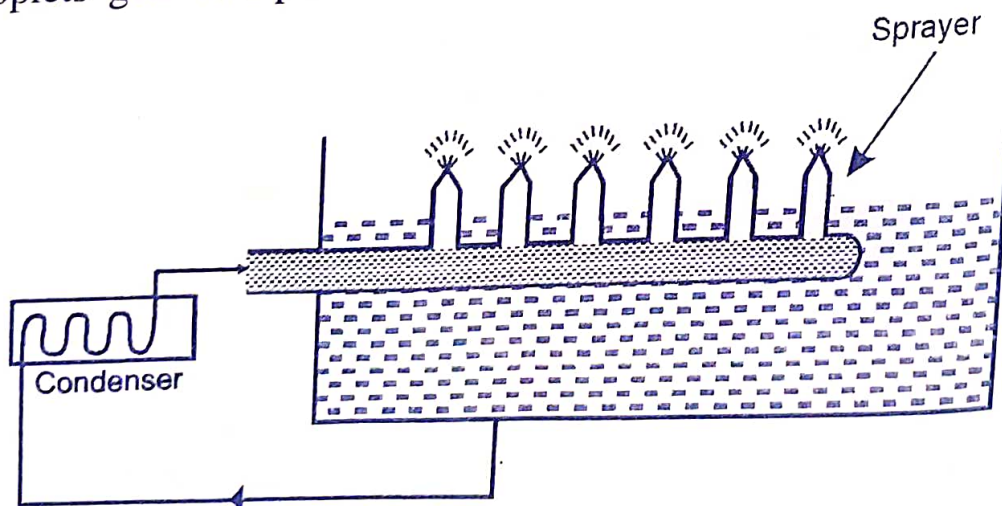


Fig. 4.7 Spray pond

4. Artificial lakes: Artificial lakes are man made bodies of water which offer possible alternative to once-through cooling. The heated effluents can be discharged into the lake at one

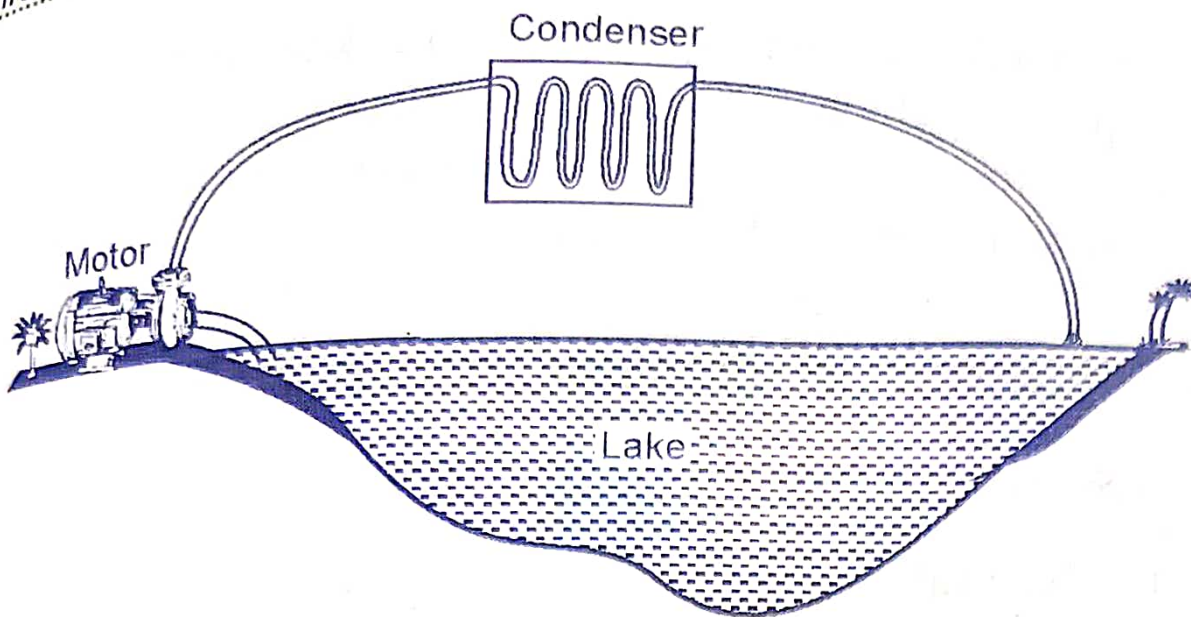


Fig. 4.8 Artificial lake

end and the water for cooling purposes may be withdrawn from the other end. The heat is eventually dissipated through evaporation.